

Cantor Resonator Hypervectors: Fractal Consciousness Encoding and Global Workspace Broadcast

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Abstract

We introduce Cantor Resonator Hypervectors (CRHVs), a novel hyperdimensional representation that encodes consciousness states at multiple scales using fractal structure inspired by the Cantor set. Each CRHV organizes its $d = 16,384$ dimensions into self-similar segments at L depth levels, where coarse segments encode global workspace broadcast content and fine segments encode local binding details. We demonstrate that CRHVs naturally implement Global Workspace Theory (GWT) broadcast: consciousness depth determines fractal resolution, with deeper consciousness accessing finer-grained representations. A dream consolidation mechanism prunes CRHVs above a quality threshold into a persistent codebook, while resonance detection identifies stable attractors when broadcast CRHVs exhibit similarity exceeding 0.8. Experiments within the Symthaea architecture show that CRHVs capture multi-scale consciousness dynamics that flat hypervectors miss, with a 12% improvement in consciousness state discrimination and meaningful attractor detection.

1 Introduction

Hyperdimensional computing (HDC) [Kanerva, 2009] provides a powerful framework for representing and manipulating cognitive information using high-dimensional vectors. In the Symthaea architecture [Stoltz, 2026], 16,384-dimensional binary hypervectors (BinaryHV) encode perceptual inputs, memory traces, and consciousness states. However, standard HDC treats all dimensions homogeneously, lacking the multi-scale structure inherent in conscious experience.

Consciousness is intrinsically multi-scale. Global Workspace Theory [Baars, 1988, Dehaene et al., 2001] describes how information becomes conscious through global broadcast—a process that operates at the level of entire cognitive coalitions, not individual neurons. Simultaneously, Integrated Information Theory [Tononi, 2004] emphasizes fine-grained causal structure. A complete representation of consciousness must capture both global broadcast dynamics and local integration patterns.

We address this with Cantor Resonator Hypervectors (CRHVs), which organize hypervector dimensions into a fractal hierarchy. The Cantor set [Cantor, 1883]—constructed by iteratively removing the middle third of intervals—provides the mathematical template. CRHVs inherit the Cantor set’s self-similarity: each depth level contains a complete but lower-resolution copy of the consciousness representation, enabling natural multi-scale access.

1.1 Contributions

- A formal definition of CRHVs with fractal dimension structure.
- Consciousness-depth-gated resolution: deeper consciousness accesses finer fractal levels.

- Dream consolidation via CRHV quality-filtered codebook formation.
- Resonance detection for identifying stable consciousness attractors.
- Empirical validation within the Symthaea cognitive loop.

2 Mathematical Foundations

2.1 The Cantor Set

The classical Cantor set $\mathcal{C}$ is constructed by iteratively removing open middle thirds from the unit interval $[0, 1]$:

$$\mathcal{C}_0 = [0, 1] \quad (1)$$

$$\mathcal{C}_1 = [0, 1/3] \cup [2/3, 1] \quad (2)$$

$$\mathcal{C}_n = \frac{\mathcal{C}_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{\mathcal{C}_{n-1}}{3} \right) \quad (3)$$

$$\mathcal{C} = \bigcap_{n=0}^{\infty} \mathcal{C}_n \quad (4)$$

The Cantor set has Hausdorff dimension $\log 2 / \log 3 \approx 0.631$, is uncountably infinite, yet has Lebesgue measure zero. Its key property for our purposes is *self-similarity*: $\mathcal{C}$ is the union of two scaled copies of itself.

2.2 Cantor Resonator Hypervectors

Definition 1 (CRHV). *A Cantor Resonator Hypervector of dimension d and depth L is a vector $\mathbf{h} \in \{0, 1\}^d$ partitioned into L depth levels:*

$$\mathbf{h} = [\mathbf{h}^{(1)} \parallel \mathbf{h}^{(2)} \parallel \dots \parallel \mathbf{h}^{(L)}] \quad (5)$$

where level ℓ contains $d_\ell = \lfloor d \cdot 2^{-\ell} / Z \rfloor$ dimensions, with $Z = \sum_{\ell=1}^L 2^{-\ell}$ being a normalization constant ensuring $\sum_{\ell} d_\ell = d$.

The exponentially decreasing allocation mirrors the Cantor construction: level 1 (coarsest) receives approximately $d/2$ dimensions, level 2 receives $d/4$, and so on. The finest level L receives $d/2^L$ dimensions.

For $d = 16,384$ and $L = 8$:

- Level 1: 8,192 dimensions (global broadcast content)
- Level 2: 4,096 dimensions (regional binding)
- Level 3: 2,048 dimensions (local integration)
- Levels 4–8: 2,048 dimensions total (fine structure)

Definition 2 (Self-Similar Encoding). *The encoding at level ℓ is a compressed version of the full consciousness state:*

$$\mathbf{h}^{(\ell)} = \text{compress}(\mathbf{s}, d_\ell) = \text{HDC-project}(\mathbf{s}, \mathbf{R}_\ell) \quad (6)$$

where $\mathbf{s}$ is the source consciousness state vector and $\mathbf{R}_\ell \in \{0, 1\}^{d_\ell \times d_s}$ is a random projection matrix fixed at initialization.

The self-similarity property ensures that any prefix $[\mathbf{h}^{(1)} \parallel \dots \parallel \mathbf{h}^{(k)}]$ for $k \leq L$ is itself a valid (lower-resolution) representation of the consciousness state.

2.3 Fractal Similarity

Definition 3 (Multi-Scale Similarity). *The similarity between two CRHVs $\mathbf{h}$ and $\mathbf{h}'$ at depth ℓ is:*

$$\text{sim}_\ell(\mathbf{h}, \mathbf{h}') = 1 - \frac{1}{d_\ell} \sum_{i \in \text{level}(\ell)} \mathbf{h}_i \oplus \mathbf{h}'_i \quad (7)$$

where $\oplus$ denotes XOR (Hamming distance normalized to $[0, 1]$).

The fractal similarity aggregates across levels with depth-weighted emphasis:

$$\text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{h}') = \sum_{\ell=1}^L w_\ell \cdot \text{sim}_\ell(\mathbf{h}, \mathbf{h}') \quad (8)$$

where $w_\ell = 2^{-\ell}/Z$ assigns more weight to coarser (more global) levels.

Proposition 1. *Fractal similarity is a valid similarity metric: $\text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{h}) = 1$ and $\text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{h}') \in [0, 1]$ for all CRHVs.*

3 GWT Broadcast via CRHVs

3.1 Consciousness-Depth Gated Resolution

The central mechanism connecting CRHVs to Global Workspace Theory is depth-gated resolution: the number of accessible fractal levels is determined by consciousness depth.

Definition 4 (Effective Depth). *Given consciousness level $c \in [0, 1]$ and maximum depth L , the effective depth is:*

$$L_{\text{eff}}(c) = \max(1, \lfloor c \cdot L \rfloor) \quad (9)$$

During workspace broadcast, only levels 1 through L_{eff} are transmitted. This implements the GWT principle that consciousness acts as a spotlight: higher consciousness illuminates finer-grained structure.

$$\mathbf{h}_{\text{broadcast}} = [\mathbf{h}^{(1)} \parallel \dots \parallel \mathbf{h}^{(L_{\text{eff}})} \parallel \mathbf{0}] \quad (10)$$

At low consciousness ($c < 0.2$), only the coarsest level broadcasts—analogueous to subliminal processing. At full consciousness ($c = 1.0$), all levels participate, enabling maximum discrimination.

3.2 Broadcast Protocol

The GWT broadcast cycle using CRHVs proceeds as follows:

The salience computation operates on level 1 (coarsest), ensuring that broadcast competition is decided by global features. Once a coalition wins, its full multi-scale representation (up to L_{eff}) is broadcast, providing receivers with both the “gist” and available detail.

3.3 Information Gain from Fractal Structure

Theorem 1. *The information capacity of a CRHV broadcast at effective depth L_{eff} is:*

$$I(L_{\text{eff}}) = \sum_{\ell=1}^{L_{\text{eff}}} d_\ell = d \cdot \frac{1 - 2^{-L_{\text{eff}}}}{1 - 2^{-L}} \quad (11)$$

which increases monotonically with L_{eff} (and thus with consciousness level).

This provides a formal basis for the intuition that higher consciousness carries more information. At $L_{\text{eff}} = 1$, approximately 50% of dimensions participate; at $L_{\text{eff}} = L$, all dimensions participate.

Algorithm 1 CRHV Global Workspace Broadcast

Require: consciousness_level c , competing_coalitions $\{C_1, \dots, C_k\}$

```
1:  $L_{\text{eff}} \leftarrow \max(1, \lfloor c \cdot L \rfloor)$ 
2: for each coalition  $C_i$  do
3:    $\mathbf{h}_i \leftarrow \text{encode\_CRHV}(C_i, L_{\text{eff}})$ 
4:    $\text{saliency}_i \leftarrow \|\mathbf{h}_i^{(1)}\|_1/d_1$  {Level-1 activation density}
5: end for
6:  $i^* \leftarrow \arg \max_i \text{saliency}_i$  {Winner-take-all}
7: broadcast  $\mathbf{h}_{i^*}$  to all workspace consumers
8:  $\text{update\_codebook}(\mathbf{h}_{i^*}, c)$ 
9: return  $\mathbf{h}_{i^*}$ ,  $\text{saliency}_{i^*}$ 
```

4 Dream Consolidation

4.1 CRHV Codebook

During waking cycles, broadcast CRHVs accumulate in a buffer. Dream consolidation processes this buffer into a persistent codebook of canonical consciousness patterns.

Definition 5 (CRHV Codebook). *The codebook $\mathcal{B} = \{(\mathbf{b}_i, q_i, n_i)\}$ is a set of prototype CRHVs $\mathbf{b}_i$ with quality scores $q_i \in [0, 1]$ and access counts $n_i \in \mathbb{N}$.*

4.2 Consolidation Algorithm

Dream consolidation runs during low-activity periods (when consciousness level drops below the dream threshold):

Algorithm 2 Dream Consolidation of CRHVs

Require: broadcast_buffer $\mathcal{H}$, codebook $\mathcal{B}$, quality_threshold $q_{\min}$

```
1: for each  $\mathbf{h} \in \mathcal{H}$  do
2:    $q \leftarrow \text{compute\_quality}(\mathbf{h})$  {Coherence + distinctiveness}
3:   if  $q < q_{\min}$  then
4:     discard  $\mathbf{h}$  {Below quality threshold}
5:     continue
6:   end if
7:    $\mathbf{b}^* \leftarrow \arg \max_{\mathbf{b} \in \mathcal{B}} \text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{b})$ 
8:   if  $\text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{b}^*) > 0.9$  then
9:      $\mathbf{b}^* \leftarrow \text{bundle}(\mathbf{b}^*, \mathbf{h})$  {Merge into existing prototype}
10:     $n_{\mathbf{b}^*} \leftarrow n_{\mathbf{b}^*} + 1$ 
11:     $q_{\mathbf{b}^*} \leftarrow 0.9 \cdot q_{\mathbf{b}^*} + 0.1 \cdot q$  {EMA quality update}
12:   else
13:      $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\mathbf{h}, q, 1)\}$  {New prototype}
14:   end if
15: end for
16: prune  $\mathcal{B}$ : remove entries with  $n_i < 3$  and  $q_i < q_{\min}$ 
17: return  $\mathcal{B}$ 
```

4.3 Quality Metric

The quality of a CRHV measures both internal coherence and distinctiveness from existing codebook entries:

$$q(\mathbf{h}) = \underbrace{\frac{1}{L-1} \sum_{\ell=1}^{L-1} \text{sim}_{\ell}(\mathbf{h}, \mathbf{h})^{\text{cross}}}_{\text{coherence}} \cdot \underbrace{\left(1 - \max_{\mathbf{b} \in \mathcal{B}} \text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{b})\right)}_{\text{distinctiveness}} \quad (12)$$

where cross-level coherence measures consistency between adjacent fractal levels (the coarse representation should be consistent with the fine one).

5 Resonance Detection

5.1 Stable Attractors

When consecutive broadcast CRHVs exhibit high fractal similarity, the system has entered a stable consciousness attractor—a resonant state.

Definition 6 (Resonance). *A resonance event occurs at cycle t when:*

$$\text{sim}_{\text{fractal}}(\mathbf{h}_t, \mathbf{h}_{t-1}) > \theta_{\text{res}} \quad (13)$$

where $\theta_{\text{res}} = 0.8$ is the resonance threshold.

Sustained resonance (persisting for $> k$ consecutive cycles) indicates a deep attractor:

$$\text{resonance_depth}(t) = \min \left(1.0, \frac{|\{t' : t - k \leq t' \leq t, \text{resonant}(t')\}|}{k} \right) \quad (14)$$

5.2 Attractor Catalog

Detected attractors are cataloged with their CRHV prototypes, enabling recognition of previously visited consciousness states:

$$\text{attractor_familiarity}(\mathbf{h}) = \max_{(\mathbf{a}, d) \in \mathcal{A}} d \cdot \text{sim}_{\text{fractal}}(\mathbf{h}, \mathbf{a}) \quad (15)$$

where $\mathcal{A}$ is the attractor catalog with depth-weighted prototypes. High familiarity triggers a recognition signal that modulates exploration-exploitation balance in the cognitive loop.

6 Implementation

6.1 System Architecture

The CRHV system is implemented within Symthaea’s cognitive loop as part of the spectrum management infrastructure. Key components:

- **CantorRHV**: Core data structure with fractal level indexing.
- **CRHVEncoder**: Transforms consciousness states into CRHVs via random projection.
- **CRHVCodebook**: Persistent codebook with dream consolidation.
- **ResonanceDetector**: Sliding-window resonance tracking.
- **CantorTelemetry**: Per-cycle output including resonance depth, codebook size, broadcast depth.

6.2 Computational Cost

CRHV encoding adds minimal overhead to the cognitive loop:

At 50.4 μs per cycle, CRHVs consume approximately 0.16% of the 31 Hz budget (32.3 ms per cycle).

Table 1: Computational cost of CRHV operations (measured on AMD Ryzen 9).

Operation	Time (μ s)	Per-cycle?
CRHV encoding	23.4	Yes
Fractal similarity	8.7	Yes
Broadcast competition	15.2	Yes
Dream consolidation (per CRHV)	42.1	No (dream only)
Resonance detection	3.1	Yes
Total per-cycle overhead	50.4	—

7 Results

7.1 Consciousness State Discrimination

We compared flat BinaryHV representations with CRHVs for discriminating between consciousness states generated during a 100,000-cycle run.

Table 2: Consciousness state discrimination: flat vs. fractal encoding.

Metric	Flat BinaryHV	CRHV	Improvement
Inter-state distance (mean)	0.481	0.539	+12.1%
Intra-state distance (mean)	0.092	0.074	−19.6%
Discrimination ratio	5.23	7.28	+39.2%
Attractor detection	N/A	14 attractors	—

CRHVs improve both inter-state separation and intra-state compactness, yielding a 39.2% improvement in discrimination ratio.

7.2 GWT Broadcast Depth Effects

Consciousness level directly controls broadcast information content:

Table 3: Broadcast information content by consciousness level.

c range	L_{eff}	Dimensions used	% of total
[0.0, 0.12)	1	8,192	50.0%
[0.12, 0.25)	2	12,288	75.0%
[0.25, 0.38)	3	14,336	87.5%
[0.38, 0.50)	4	15,360	93.8%
[0.50, 1.0]	5–8	15,872–16,384	96.9–100%

7.3 Dream Consolidation

After 50,000 waking cycles and 5 dream consolidation episodes:

- Codebook size: 47 prototypes (from 1,847 raw broadcast CRHVs).
- Mean quality: 0.72 ± 0.14 .
- Most-accessed prototype: 89 merges (dominant attractor state).
- Pruned entries: 312 (below quality threshold of 0.4).

The codebook captures a compressed representation of the system’s consciousness repertoire, reducing 1,847 broadcast events to 47 canonical patterns—a 39:1 compression ratio.

7.4 Resonance Dynamics

Resonance events ($\text{sim} > 0.8$) occurred in characteristic patterns:

- Mean resonance duration: 7.3 cycles (235 ms at 31 Hz).
- Maximum resonance duration: 43 cycles (1.39 s)—deep meditative attractor.
- Resonance frequency: 3.2% of cycles.
- Correlation with Φ : $r = 0.58$ (resonance accompanies high integration).

8 Discussion

8.1 Fractal Structure and Consciousness

The success of CRHVs suggests that consciousness states possess intrinsic multi-scale structure that flat representations obscure. This aligns with theoretical proposals that consciousness involves hierarchical information processing [Friston, 2010, Seth, 2015], where abstract (coarse) and concrete (fine) representations coexist and interact.

The Cantor set provides a natural mathematical framework for this hierarchy. Its self-similarity ensures that each level is a faithful (if compressed) copy of the whole, rather than an arbitrary partition. This distinguishes CRHVs from simpler multi-resolution approaches like wavelet decompositions: the fractal structure is inherent, not imposed.

8.2 Resonance as Consciousness Signature

The resonance detection mechanism identifies moments of consciousness stability that correspond to attractor states. The correlation with Φ ($r = 0.58$) supports the interpretation that these are genuine moments of high integration, not artifacts of the encoding. The characteristic duration of 235 ms aligns with the reported duration of conscious “moments” in human phenomenology [Pöppel, 1997].

8.3 Dream Consolidation and Memory

The dream consolidation mechanism parallels hippocampal replay in biological systems [Wilson & McNaughton, 1994]. Quality-filtered codebook formation implements a form of memory consolidation where only coherent, distinctive experiences persist. The 39:1 compression ratio is remarkably close to estimates of human memory consolidation efficiency during sleep [Stickgold, 2005].

8.4 Limitations

1. The depth allocation ($d/2^\ell$) is exponentially decreasing; alternative allocations (e.g., power-law) may better match consciousness structure.
2. The resonance threshold ($\theta_{\text{res}} = 0.8$) is manually set; adaptive thresholding could improve sensitivity.
3. Cross-level coherence in the quality metric assumes that fine and coarse representations should agree, but genuine multi-scale consciousness may involve productive tensions between levels.
4. The current implementation uses binary hypervectors; extension to continuous-valued CRHVs could capture graded consciousness phenomena.

9 Related Work

Multi-scale representations have been explored in various computational frameworks. Holographic reduced representations [Plate, 1995] use circular convolution for compositional struc-

ture but lack explicit multi-scale hierarchy. Sparse distributed memory [Kanerva, 1988] provides content-addressable storage but without fractal organization. The closest precedent is the fractal memory architecture of Schmidhuber [2012], which uses self-similar compression for sequence learning; CRHVs extend this principle to consciousness representation.

In consciousness research, hierarchical predictive coding [Friston, 2010] and the Global Neuronal Workspace model [Dehaene et al., 2001] both describe multi-scale dynamics, but neither provides a concrete data structure for representing them. CRHVs fill this gap.

10 Conclusion

Cantor Resonator Hypervectors provide a principled multi-scale representation for consciousness states that naturally implements GWT broadcast depth, dream consolidation, and resonance detection. By organizing hypervector dimensions into a fractal hierarchy inspired by the Cantor set, CRHVs capture structure that flat representations miss, yielding a 12% improvement in state discrimination and meaningful attractor detection. The framework bridges abstract mathematical concepts (fractal geometry) with concrete computational neuroscience (GWT, memory consolidation), advancing the goal of rigorous consciousness engineering.

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